Solar Panels as Extreme Habitats for Fungal Biosoiling Across Sites, Seasons, and Disturbance

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PreMiEr Precision Microbiome Engineering

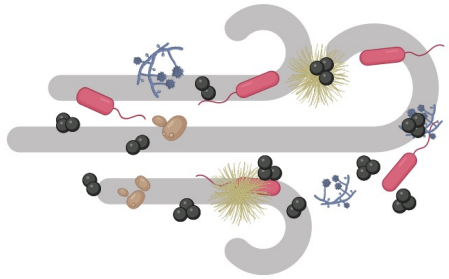
NSF Engineering Research Center

NSF

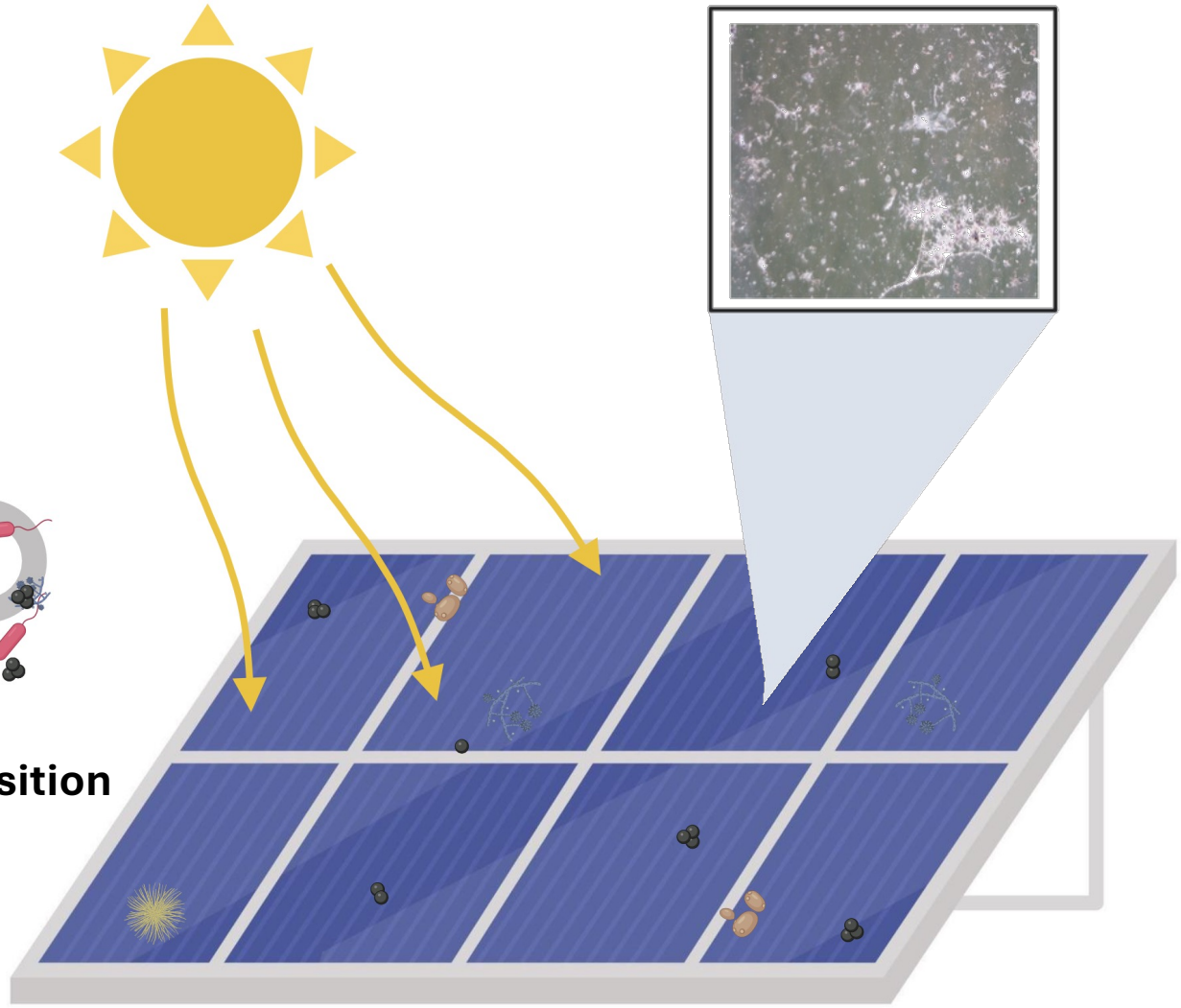
Solar Unsoiled

Duke | PRATT SCHOOL of ENGINEERING

Atmospheric deposition of fungi on solar panels



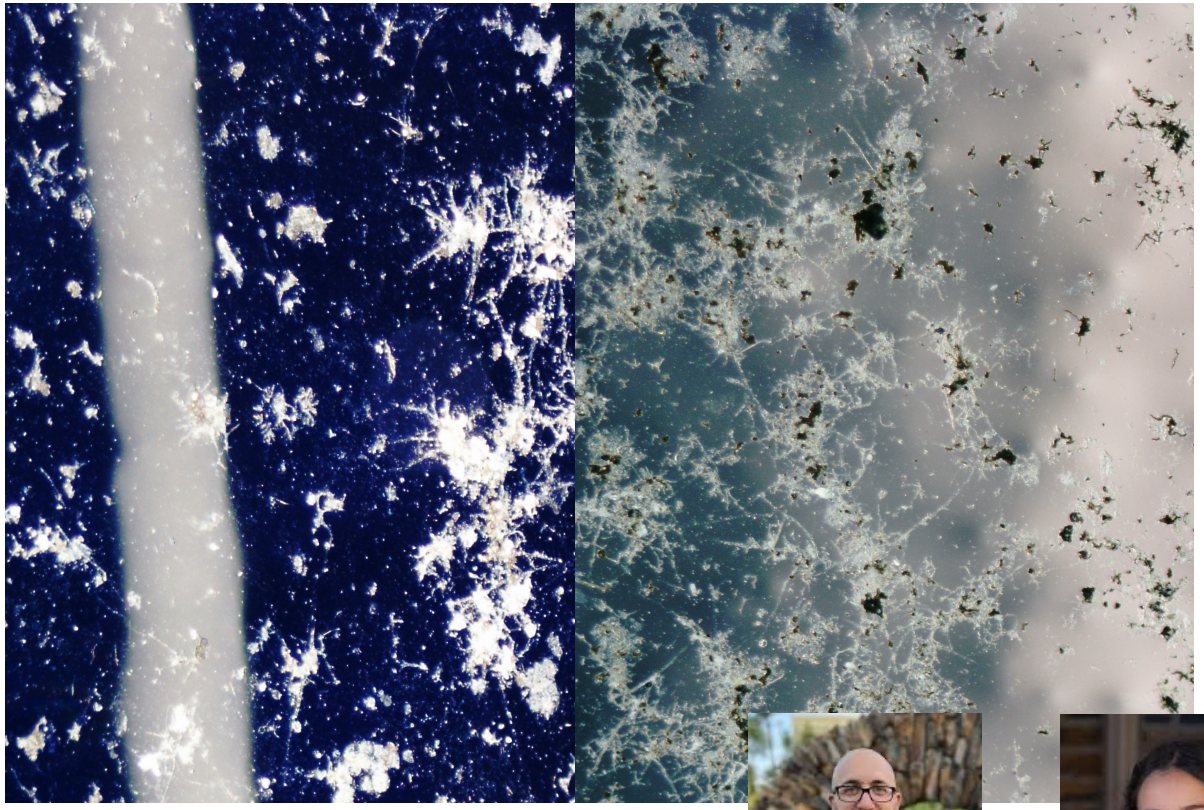
Atmospheric deposition



$330 million

in losses due to biosoiling in the subtropic region alone

Working with Solar Unsoiled to understand the factors behind biosoiling



Images taken by Solar Unsoiled

Solar Unsoiled

Project Objective

To characterize fungal colonization on solar panels, the impact across **different sites, seasons, and following disturbance**



Mike Valerino



Alexandra Rivera

Hypotheses



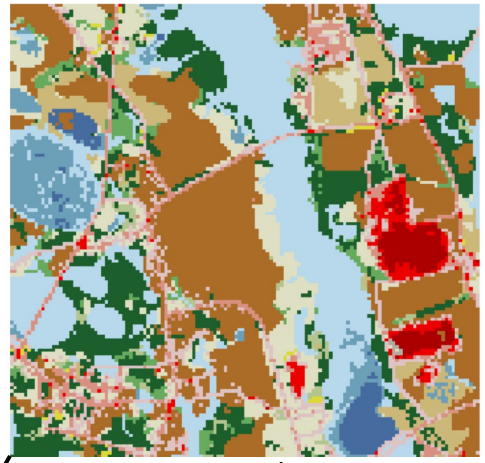
1. Community Composition *Between* Sites
The community will show dominance of black fungi, but will show site specific assemblages



2. Community Composition *Within* Sites
Environmental variables will influence fungal assemblages, both over a year undisturbed and during recolonization

Sampling three sites across North Carolina

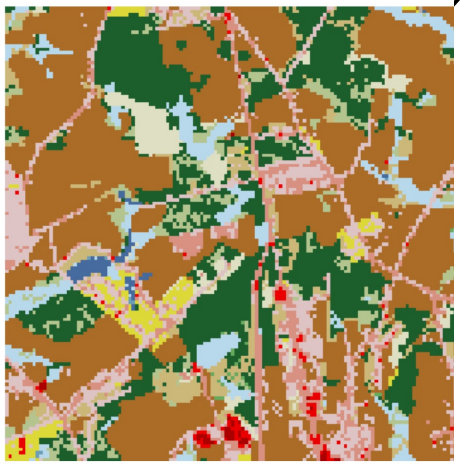
Hope Mills NC



March 24th, 2024

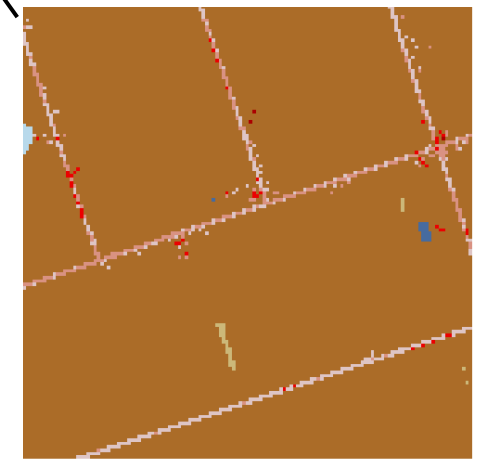


Laurinburg NC



January 24th, 2024

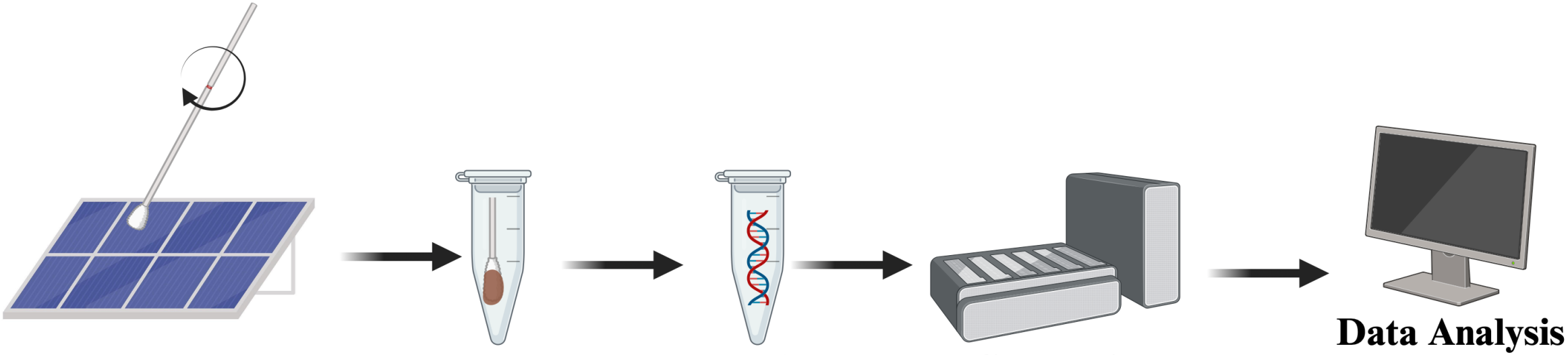
Wilkinson NC



October 8th, 2024



Long read sequencing of samples



Sampling
PBS Soaked Swab

DNA Extraction
DNEasy Soil Power Pro Kit

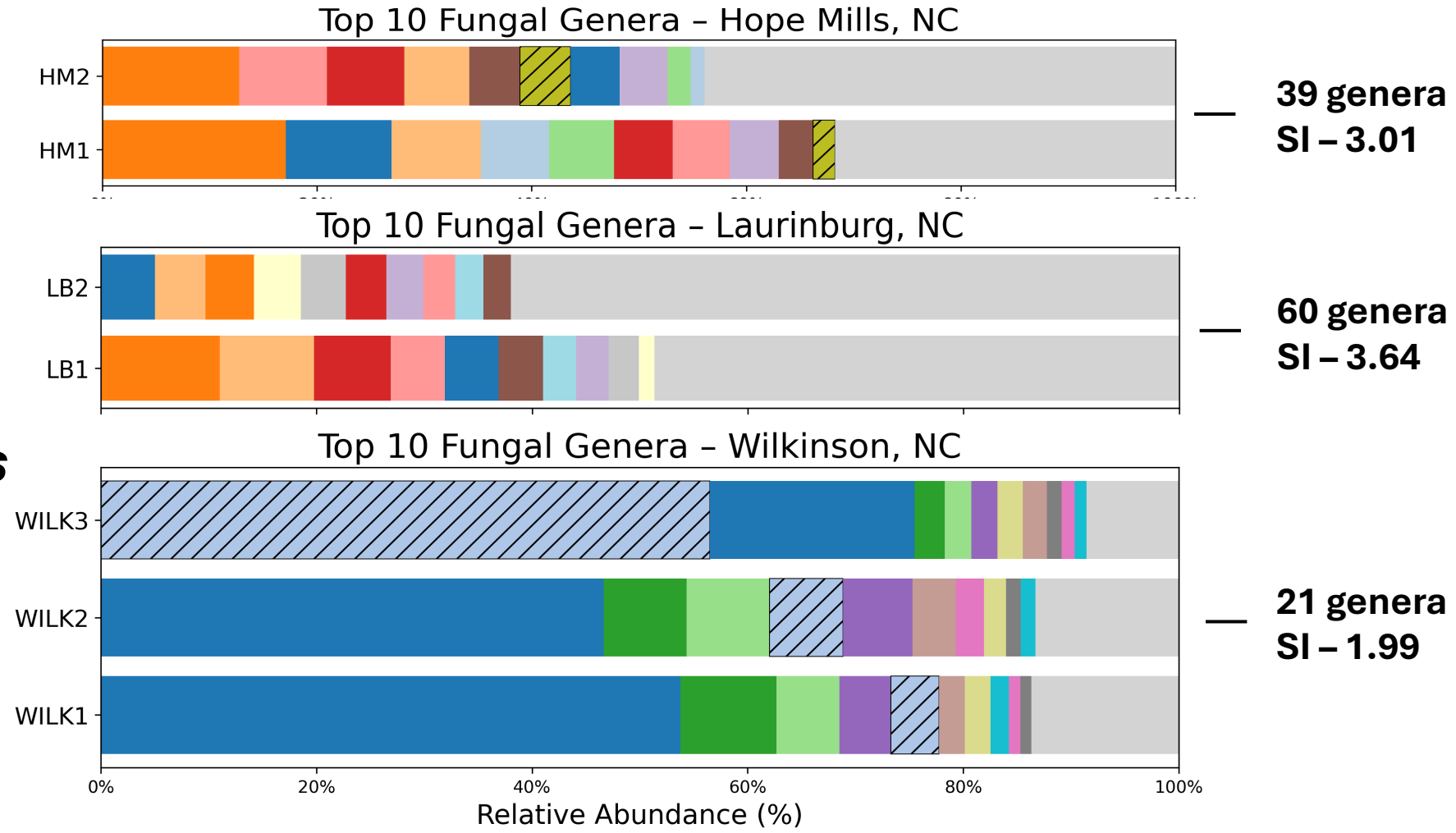
Sequencing
PromethION

Data Analysis
Several online pipelines

Fungal genera differed between sites

Most fungi are within the black fungal class ***Dothideomycetes***

NCBI nt database for alignment and MEGAN for classification

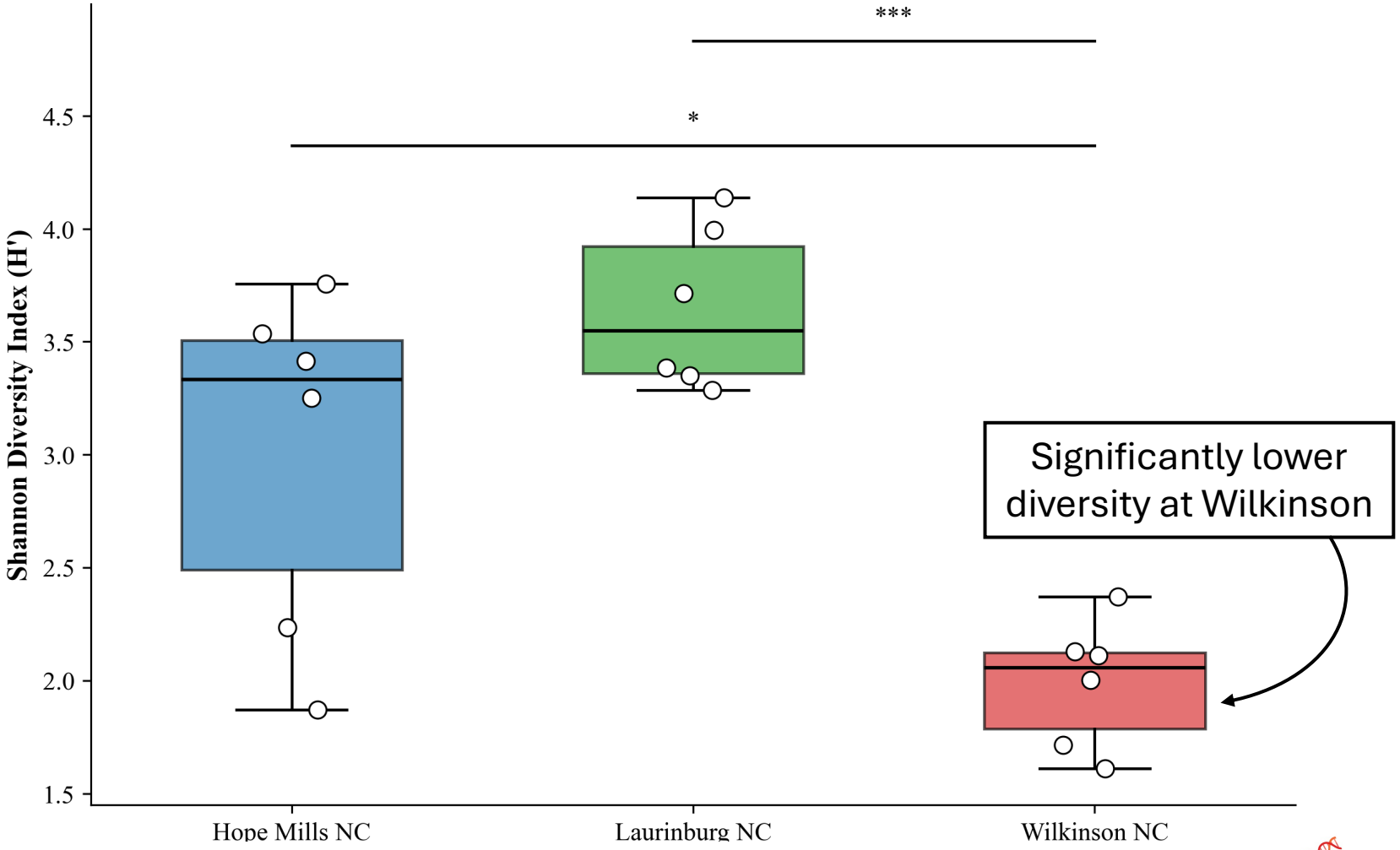


Non-melanized

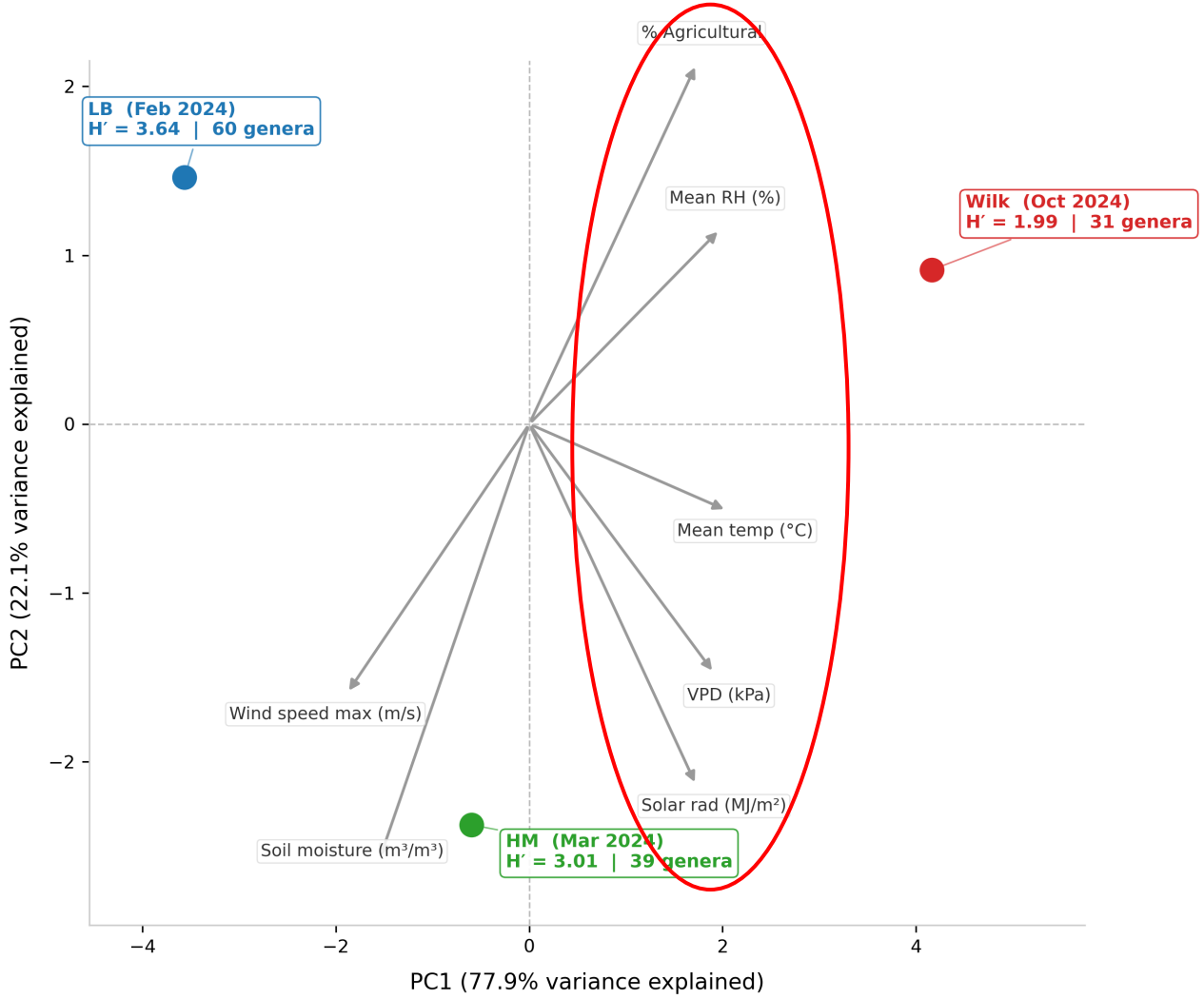
SOLID — Melanized

Shannon index showed site specific trends

Shannon index (H) used to measure diversity in a community by combining species richness and evenness



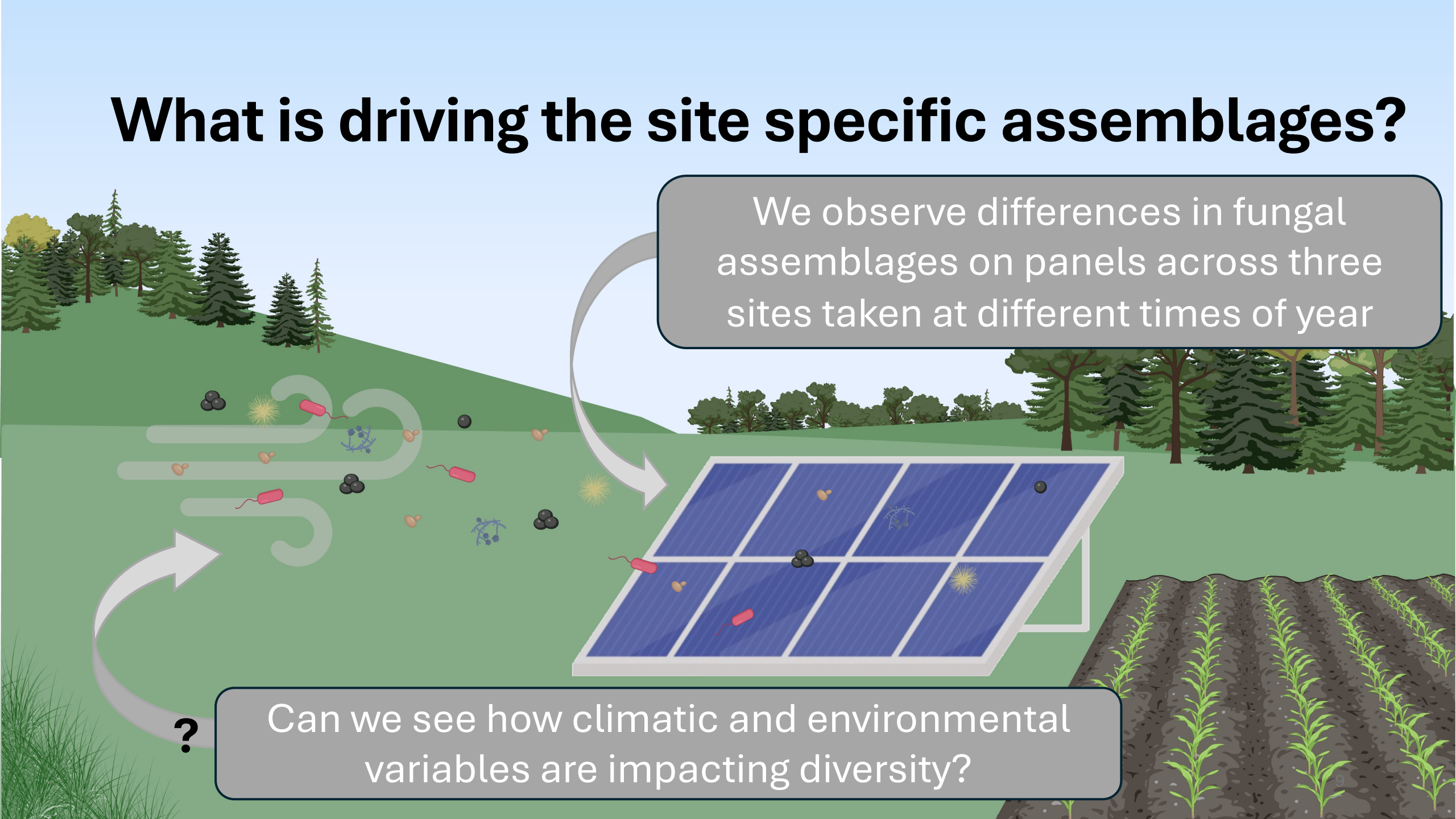
Climate and environmental variables related to shifts in diversity



Wilkinson showed the lowest diversity

- Highest agriculture is surrounding land cover
- Highest temperature
- Highest solar radiation
- Lowest windspeed

What is driving the site specific assemblages?



We observe differences in fungal assemblages on panels across three sites taken at different times of year

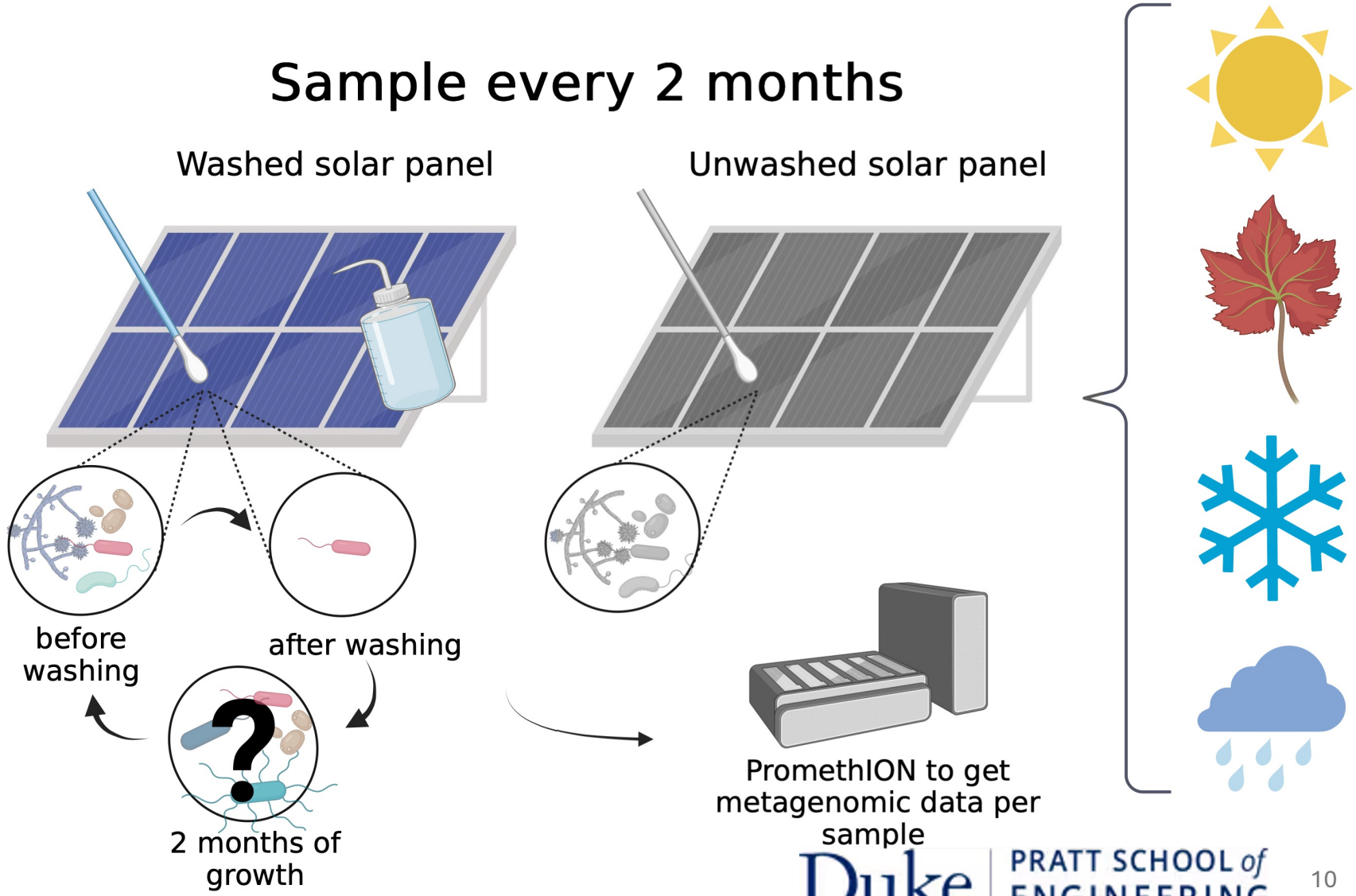
?

Can we see how climatic and environmental variables are impacting diversity?

Seasonal study to investigate diversity shifts

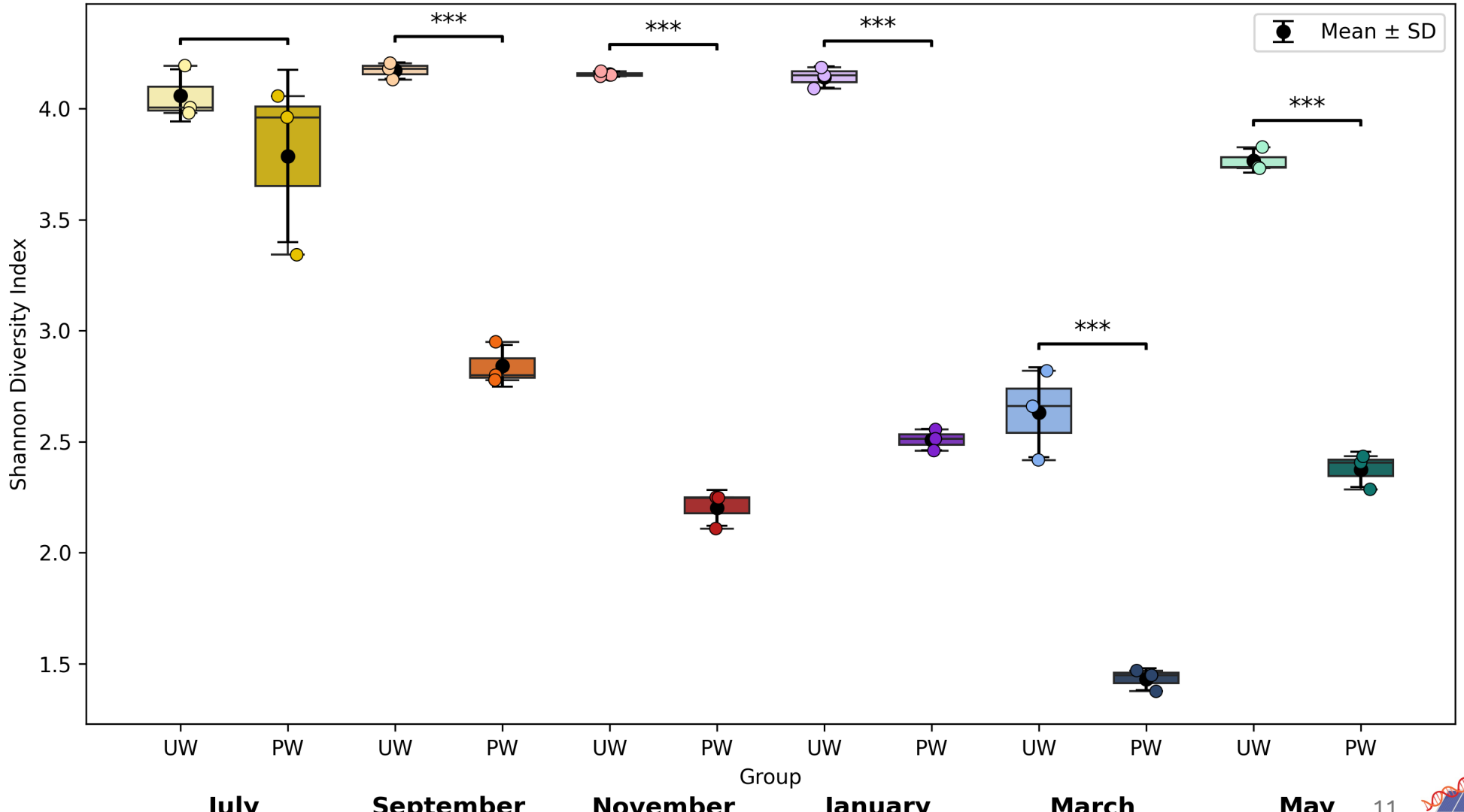
Does season affect the recolonization after cleaning

Does season change the community over a year



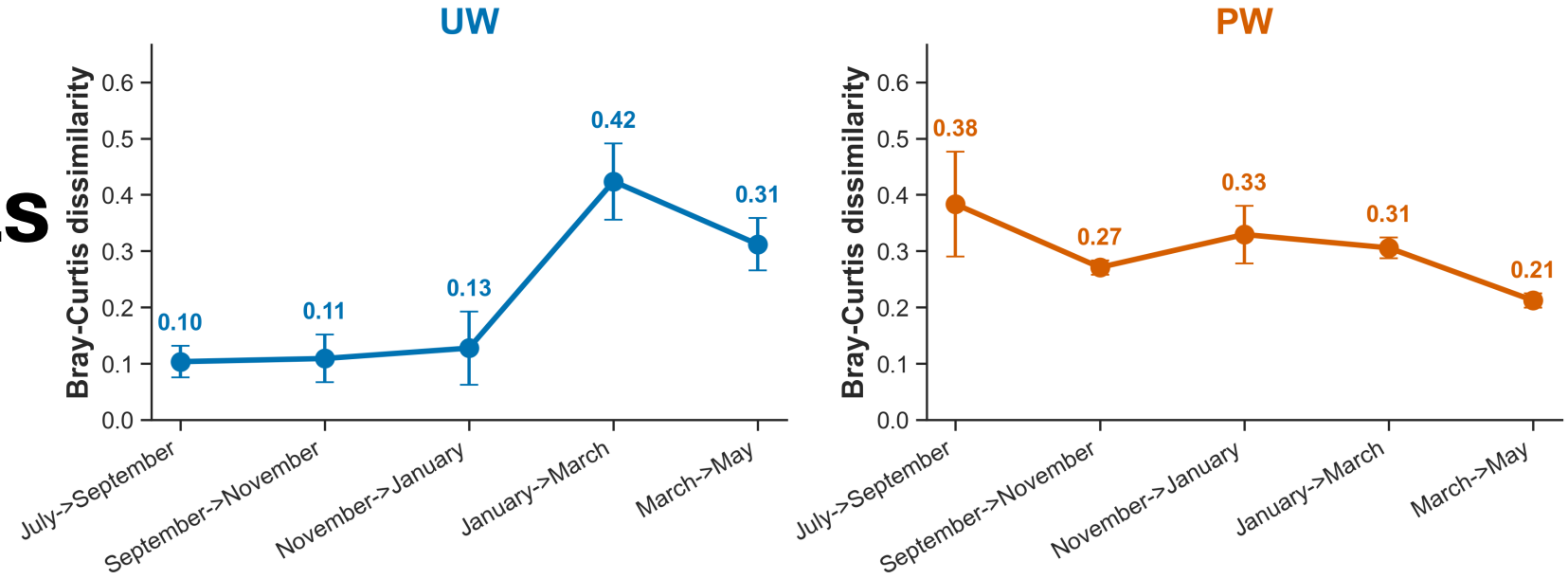
Diversity in PW decreases from UW

PW samples have significantly lower Shannon diversity compared to UW

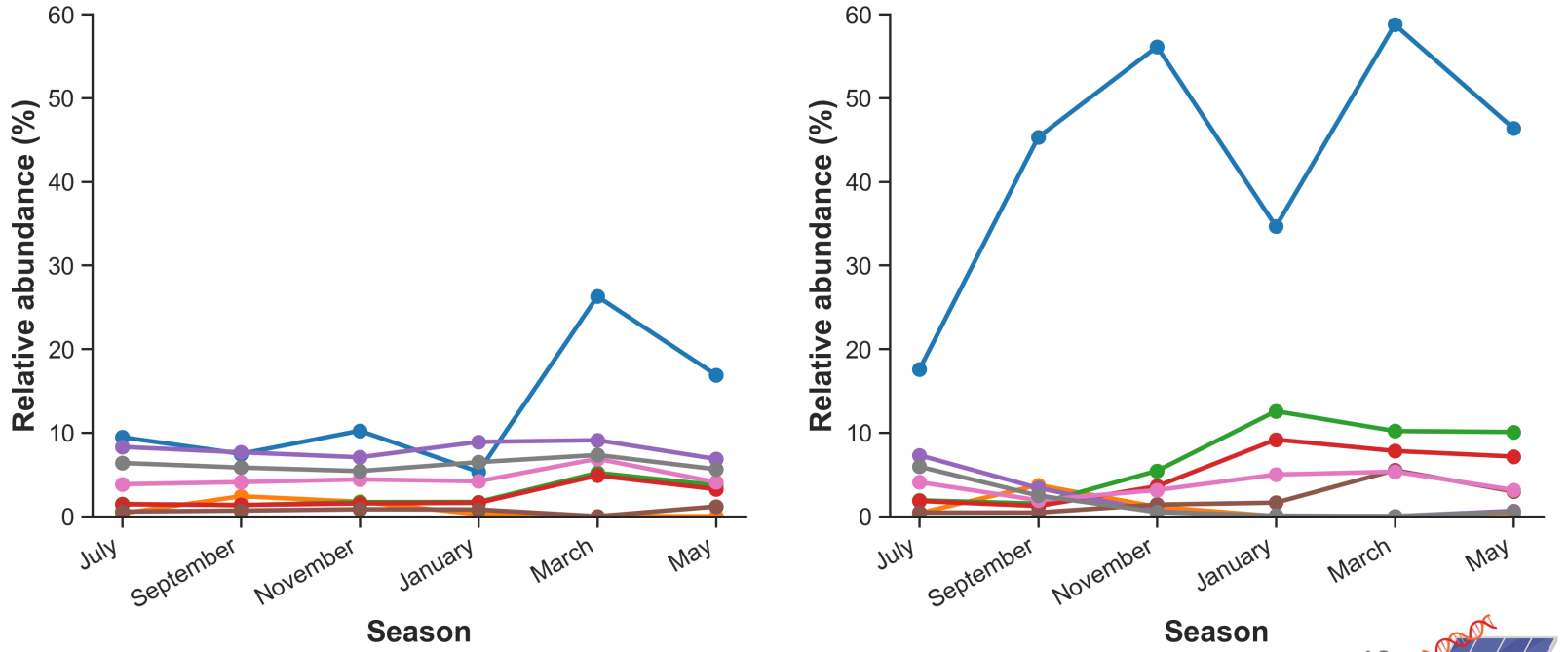


UW – Unwashed
PW – Prewashed

Alternaria is a colonizer of panels following a disturbance

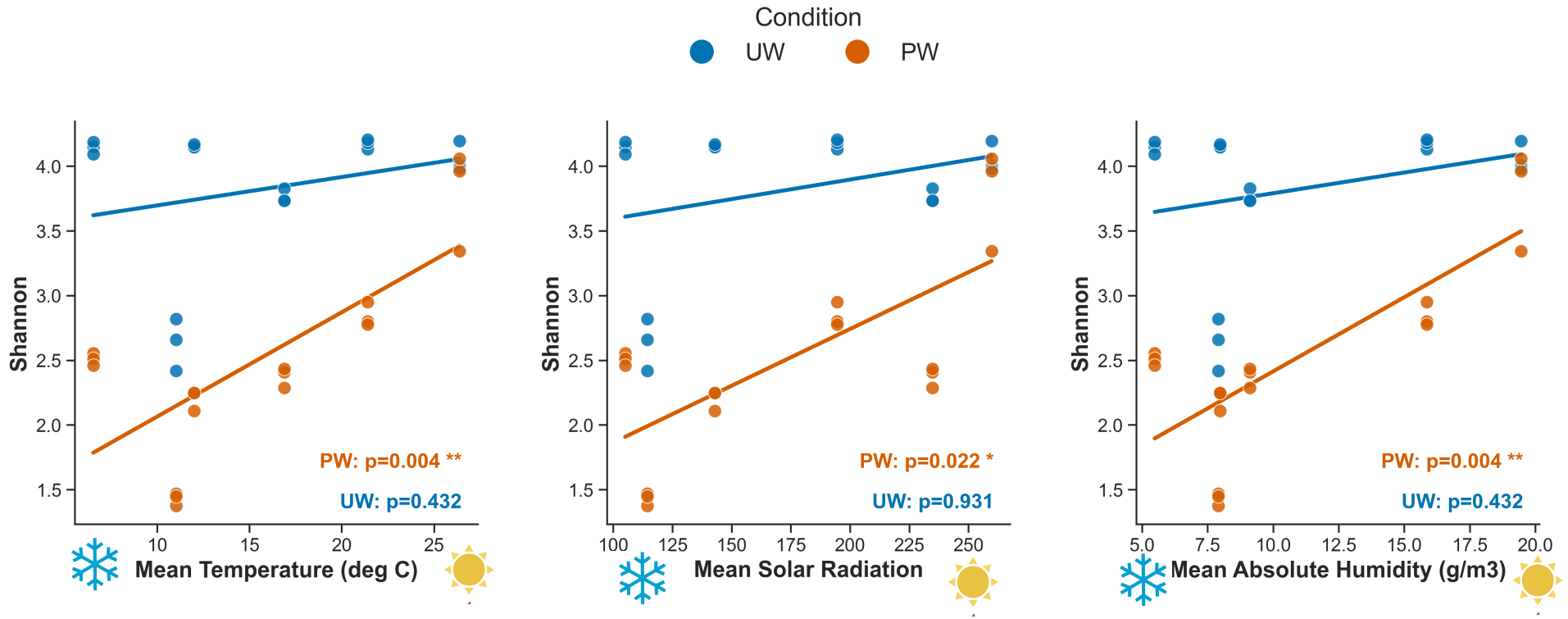




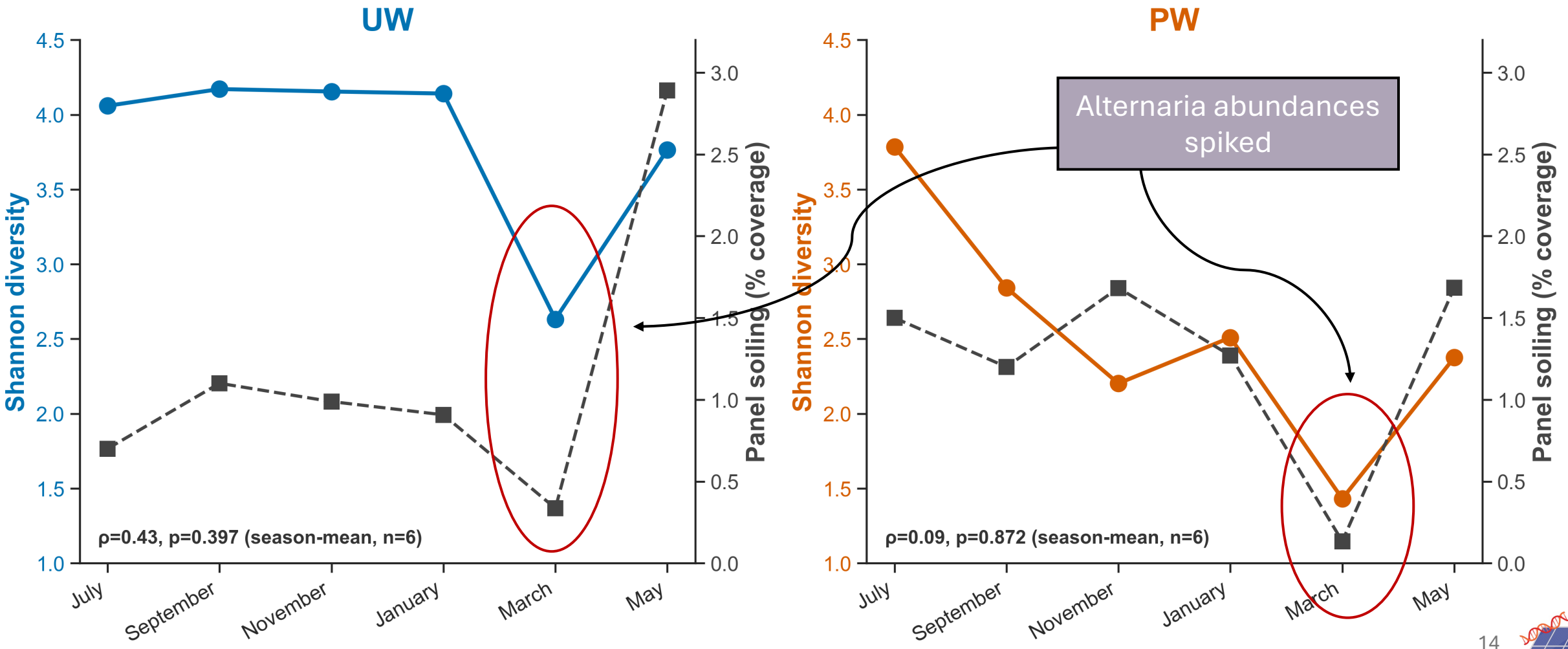


PW diversity attributed to seasonal changes

UW samples do not show the same significant relationship



Bio-soiling and diversity trends match



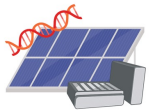
Conclusions

There are site specific differences in the fungal microbiome based on climatic and environmental variables

Washed and unwashed solar panels have distinct fungal communities

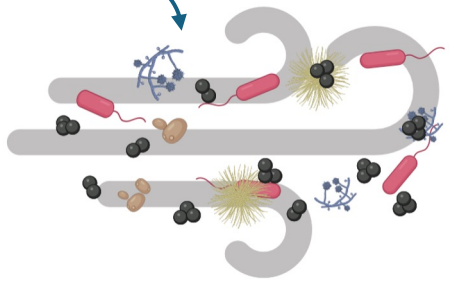
Fungal communities on washed panels have higher seasonal turnover and are more influenced by seasonal variables

A higher relative abundance of *Alternaria* could be reducing efficiency losses in solar panels

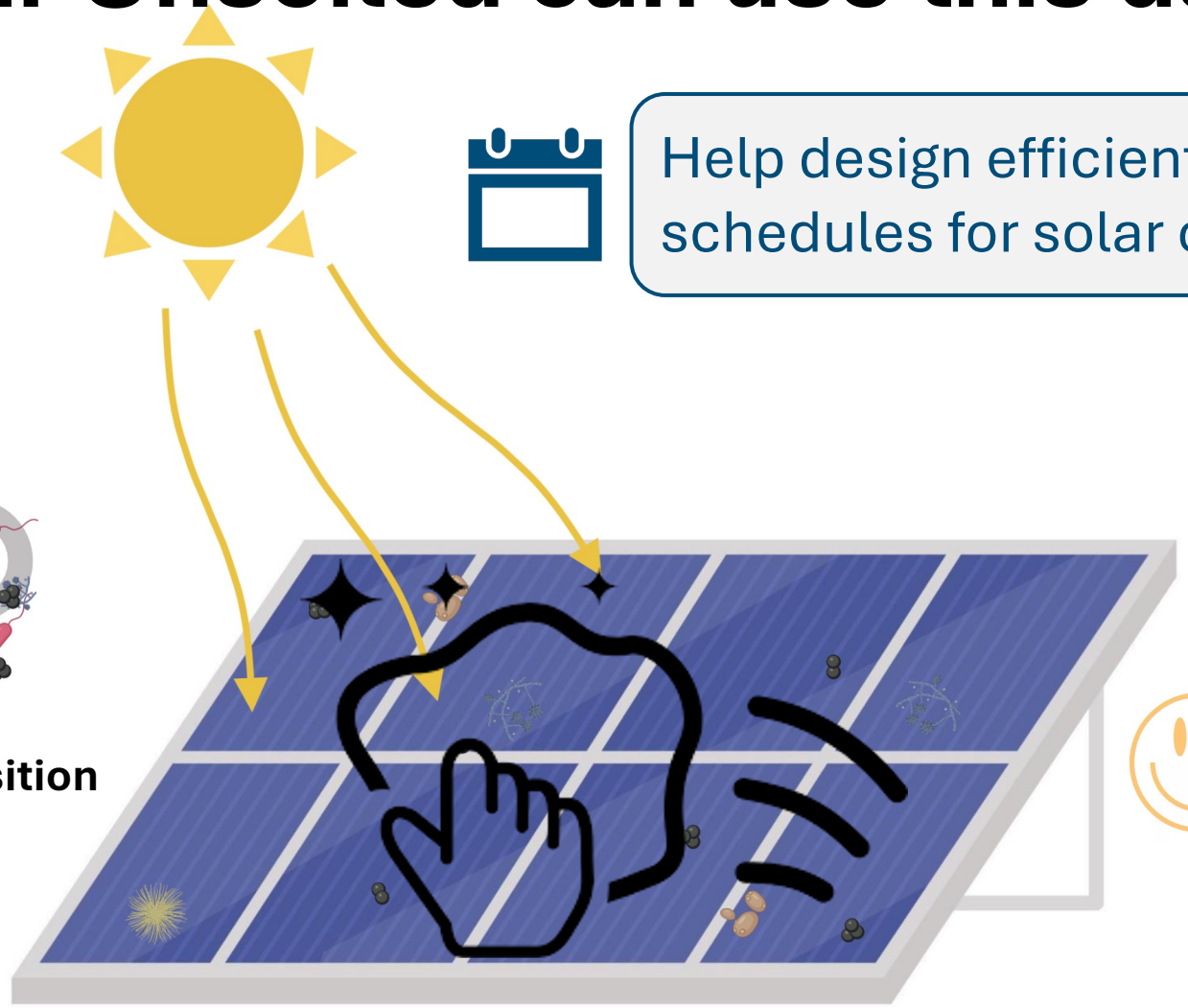


How Solar Unsoiled can use this data

Clearer picture of what drives biosoiling



Atmospheric deposition



Help design efficient cleaning schedules for solar companies

~~$330 million~~

in losses due to biosoiling in the subtropic region alone

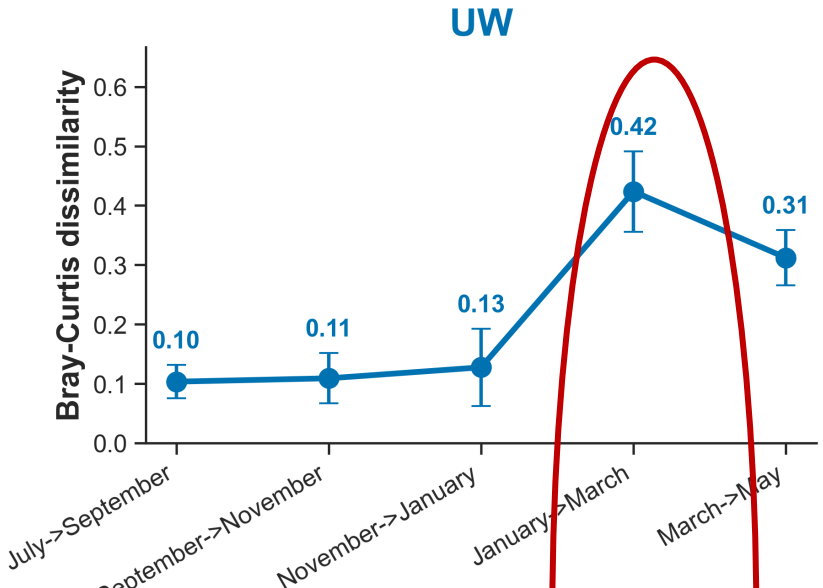
Reduce losses for their customers

Next Steps

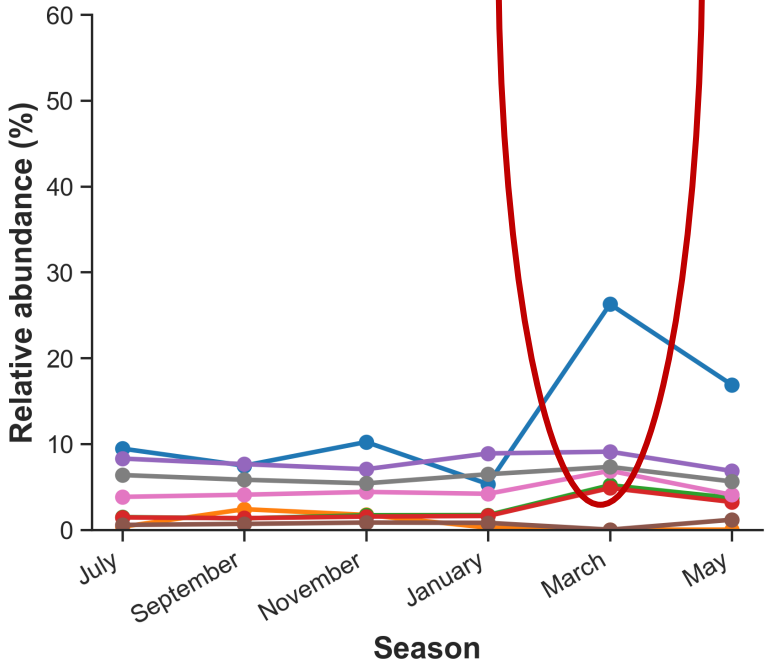
<u>Expand to more sites</u>

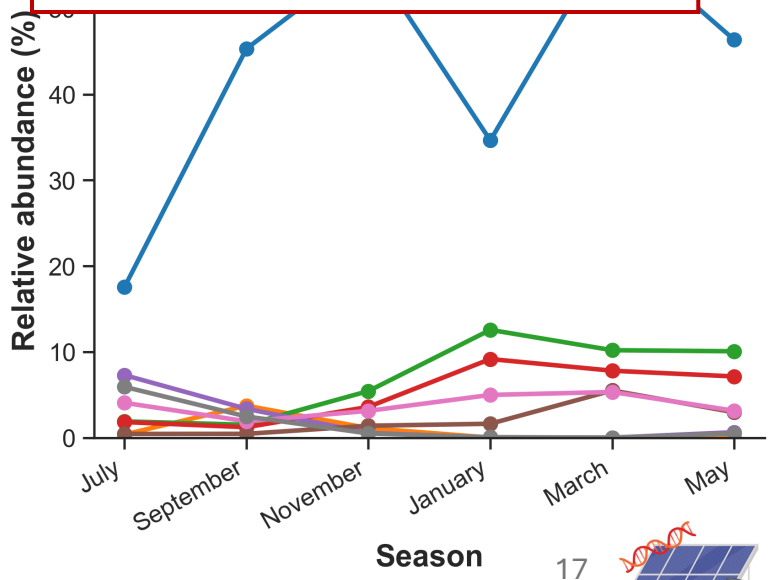
- Do we see the same shifts in March?
- How does *Alternaria* influence bio-soiling











What's happening in March?

Thank you!!!

Project Members

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Alexandra Rivera (CFO)

Dr. Mike Bergin

Dr. Andrew Jones

Matthew Muller

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